Assets of the MARISMA-IoT Case Study (Laboratory Smart Home)

Table 1. Assets of the Case Study Defined in the MARISMA-IoT Pattern			
Asset Family	**Asset Type**	**Case Study Asset**	**Model**
IoT Devices	Hardware	Embedded microcontroller of the IoT devices	Generic
	Software	Internal firmware of the IoT devices	Generic
	Sensors	Temperature and humidity sensor	Generic
	Sensors	Motion sensor	Tuya ZigBee Motion Sensor ZMIR01
	Actuators	Smart bulb	Moes E27
	Actuators	Automated blind motor	Generic
	Actuators	Smart switch	MOES ZigBee Smart Switch ZS-EUB & MOES WiFi Smart Switch ZS-EUB
	S/A (Sensor/Actuator)	Smart thermostat	MOES BHT-006GAW Series WiFi Thermostat
	S/A (Sensor/Actuator)	Smart plug	ZigBee Power Plug-001SPB2 & Athom Smart Plug PG01
IoT Ecosystem	Interface Devices	Smart display with voice assistant	Google Nest Hub 2 & Amazon Echo Show 5
	Management Devices	Smart home hub	Raspberry Pi 4 + ConBee II + Home Assistant
	Embedded Systems	Raspberry Pi mini-computer	Raspberry Pi 4
Communications	Networks	Home Wi-Fi network and Zigbee network	Generic
	Protocols	Wi-Fi protocol	Generic
	Protocols	MQTT protocol	Generic
	Protocols	ZigBee protocol	Generic
	Protocols	HTTP/HTTPS protocol	Generic
Infrastructure	Routers	Home Wi-Fi router	TP-Link Archer MR200
	Gateways	Multi-protocol gateway	SONOFF B09ZQQZSQZ
	Power Sources	Rechargeable lithium battery in compatible IoT devices	Generic
	Security Assets	WPA2 encryption on Wi-Fi network and secure credentials	Generic
Platform and Backend	Web Services	Open-source home automation software platform	Home Assistant
	Cloud Infrastructure and Services	Cloud-based AIoT platform	Tuya AIoT Cloud & AWS IoT Cloud Platform
Decision Making	Data Mining	Artificial intelligence algorithm in the cloud	Generic
	Processing and Computing	Edge computing	Generic
Applications and Services	Data Analysis and Visualization	Grafana dashboard for IoT	Generic
	Device and Network Management	Unified home automation control app	Home Assistant
	Device Usage	Context-based automation	Generic
Information	At Rest	Stored historical data	Generic
	In Transit	IoT message in transmission	Generic
	In Use	Data used in real time	Generic
Audio/Visual	Digital Streaming Systems	Smart TV	Chromecast 3